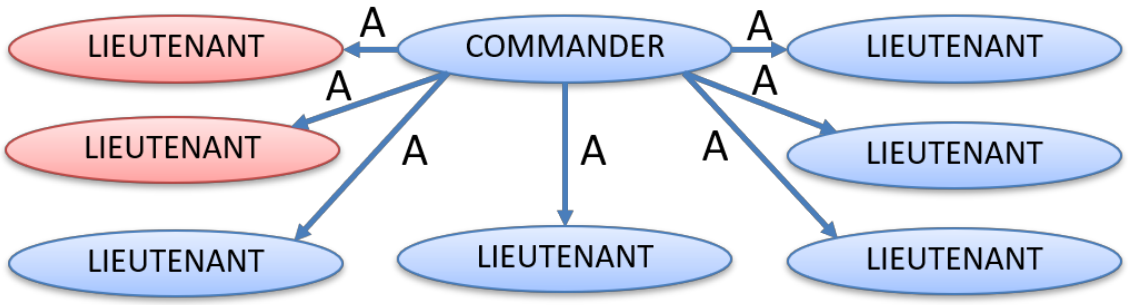
Exercise: Byzantine Fault Tolerance

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1. Imagine there are 1 loyal commander, 2 traitor lieutenants, and 5 loyal lieutenants. Please provide a concrete example where Oral Message algorithm is used among these soldiers, yet the loyal lieutenants all are able to follow the commander's command.

   You need to briefly describe the process of Oral Message algorithm in your example.

   - *The red ellipse represents the faulty nodes, while the blue one for the normal nodes*
   - *The commander sends the "Attack" order to all lieutenants to which it connects.*
   - *When each lieutenant act as a commander, a loyal lieutenant sends the original order, "Attack", while the traitor sends "Retreat".*



in this situation N = 8 (1 Commander + 7 Lieutenant) to statisfies the byzantine condition $N > 3m$ $(8 > 6)$

m = 2 (2 traitor)

The oral message process

Com = Commander

Lie = Lieutenant

1. Com send initial commands to all 7 Lie

2. Each Lie forward command to each other 6 Lie

   5 Loyal Lie will send the original command

   2 traitor Lie will send a conflicting command

3. now each Lie will receive all 7 Command

   5 original command from 1 Com + 4 Loyal Lie

   2 conflict command from 2 traitor

4. Loyal Lie will use majority vote for final action

   5 original command will over power 2 conflict command